A GERSTEN–WITT COUNTEREXAMPLE IN DIMENSION FIVE

GPT-6 ASTRA DIRECTED BY FANGZHOU JIN

Abstract. Based on recent counter-examples to the Gersten conjecture with integral, rational and finite coeffcients by Feld, Guo and OpenAI, we propose a counter-example to the Gersten conjecture for Witt groups.

1. Introduction

Based on recent counter-examples to the Gersten conjecture with integral, rational and finite coeffcients ([2], [3], [5], [8], [9]), we propose a counter-example to the Gersten conjecture for Witt groups.

More precisely, we formulate a Gersten–Witt counterexample in dimension five using a self-dual two-term complex associated with a matrix factorization of the element 3 in a ramified regular local ring. Its residue at a codimension-three prime is the nontrivial rank-one maximal Cohen–Macaulay class over the A_1-quadric cone $uv = x^2$. The class is a rank-one symmetric form, hence nonzero in the integral Witt group; it also survives with mod-2 coefficients. The determinant of the underlying MCM module is 2-torsion in the class group, but the Witt class itself is not 2-torsion.

1.1. The Gersten conjecture for Witt groups.

Let $(R, \mathfrak{m}, k)$ be a noetherian regular local ring of dimension d, with fraction field K. Throughout the classical integral statements, assume

$$2 \in R^{\times}.$$

In mixed characteristic $(0, p)$ this means $p \neq 2$. Let $\mathrm{W}(R)$ be the Witt group of nonsingular symmetric bilinear forms modulo metabolic forms. Since 2 is invertible, this is also the usual quadratic-form Witt theory.

The conjecture asserts exactness of the augmented complex

$$0 \longrightarrow \mathrm{W}(R) \longrightarrow \mathrm{W}(K) \longrightarrow \bigoplus_{x \in (\operatorname{Spec} R)^{(1)}} \mathrm{W}(\kappa(x)) \longrightarrow \cdots \longrightarrow \mathrm{W}(k) \longrightarrow 0. \tag{1}$$

We suppress one-dimensional coefficient twists: canonically, the codimension-a term uses the duality line $\operatorname{Ext}^a_{R_x}(\kappa(x), R_x)$, or the corresponding line in an equivalent duality convention. Trivializing these lines gives the displayed notation. The differentials are Witt residues with the appropriate transfer maps.

Exactness includes injectivity at the generic point, extension of unramified classes, and vanishing of all positive-degree cohomology of the Gersten–Witt complex. These are stronger claims than generic injectivity alone.

1.2. Summary of established cases.

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Hypothesis	Conclusion and reference
R a DVR, residue characteristic different from 2	Full exactness; also a special case of [1].
$\dim R \leq 4$, with no unramifiedness hypothesis	Full exactness, Balmer–Walter [1, Corollary 10.4].
Mixed characteristic $(0,p)$, $p \neq 2$, and $p \notin \mathfrak{m}^2$	Full exactness in every dimension, Jacobson [6].
R essentially smooth over a DVR with 2 invertible	Full exactness, including hermitian versions, Gille–Panin [4].
R geometrically regular over such a DVR	Full exactness by the Popescu extension of [4].
Arbitrary regular local R with $2 \in R^\times$, after tensoring (13) with $\mathbb{Z}[1/2]$	Exactness without an unramifiedness hypothesis; see [7, Remark 3.7].

1.3. **Discrete valuation rings and low dimension.** For a DVR V with fraction field F, residue field k of characteristic different from two, and uniformizer π, the sequence is

$$0 \longrightarrow \mathrm{W}(V) \longrightarrow \mathrm{W}(F) \xrightarrow{\partial_\pi} \mathrm{W}(k) \longrightarrow 0.$$

On one-dimensional forms, for $u \in V^\times$ and $r \in \mathbb{Z}$,

$$\partial_\pi \langle u\pi^r \rangle = \begin{cases} 0, & r \text{ even}, \\ \langle \bar{u} \rangle, & r \text{ odd}. \end{cases}$$

The choice of π trivializes the residue line. Lifting units shows surjectivity directly. Exactness identifies its kernel with forms extending over V. Completeness and henselianity are not required.

More generally, Balmer–Walter prove (13) for every regular local ring of dimension at most four with 2 invertible. Their argument uses the Gersten–Witt spectral sequence and its four-periodic pattern [1]. Thus ramified mixed-characteristic regular local rings in dimensions two, three, and four are included.

1.4. **Unramified rings and regular morphisms over DVRs.** In mixed characteristic $(0,p)$ a regular local ring is called unramified here when

$$p \notin \mathfrak{m}^2, \quad \text{equivalently, } R/pR \text{ is regular.}$$

Jacobson proves the integral Gersten–Witt conjecture in this case for $p \neq 2$, without a dimension restriction [6]. The result is Theorem 2.8 in the linked author manuscript and is cited as Theorem 3.8 in the published version. The proof relates the fundamental-ideal filtration and signature methods to Gersten results for etale cohomology.

Gille–Panin prove exactness for local rings essentially smooth over a DVR, and extend it to local rings geometrically regular over a DVR using Popescu's theorem [4]. A regular morphism here is flat with geometrically regular fibers; regularity of the source ring alone is not sufficient.

The base DVR may itself be ramified over the prime integers. For example, for an odd prime p and $e \geq 2$, let

$$V = \mathbb{Z}_p[\pi]/(\pi^e - p), \qquad R = V[t_1, \ldots, t_r]_{(\pi, t_1, \ldots, t_r)}.$$

Then R is essentially smooth over V, so the theorem applies, although $p = \pi^e \in \mathfrak{m}_R^2$. This illustrates why "unramified over the integers" and "geometrically regular over some DVR" are different hypotheses.

1.5. **Hermitian extensions.** The Gille–Panin theorem also covers Azumaya algebras with involution of the first or second kind, in the same geometric setting and with 2 invertible [4]. Its specialization to the algebra R with identity involution gives the classical theorem above. One should not transfer every classical low-dimensional theorem to arbitrary Azumaya algebras without checking additional hypotheses. Likewise, Witt groups and higher Grothendieck–Witt groups are different invariants; the exactness statements here are for the specified Gersten–Witt complexes.

1.6. **Inverting two and the remaining scope.** After tensoring the classical complex with $\mathbb{Z}[1/2]$, Gersten exactness holds for arbitrary regular local rings in the stated 2-invertible setting. Jacobson established an excellent case; the excellence restriction was subsequently removed. The precise references are recorded in [7, Remark 3.7].

It follows that any integral defect remaining in this setting is 2-primary torsion: each element of the defect is annihilated by some power of 2. This does not assert a uniform exponent. Inverting 2 in the coefficient groups is distinct from the standing assumption that 2 is already a unit in the base ring.

The theorems listed here do not settle the unrestricted integral case of ramified regular local rings of dimension at least five without an appropriate geometric regularity hypothesis. This note does not claim a counterexample in that range. In particular, a K-theory counterexample over a three-dimensional regular local ring with 2 invertible cannot be a classical Witt–Gersten counterexample over the same ring: the low-dimensional theorem already excludes it.

In mixed characteristic $(0, 2)$, the classical hypotheses above fail. Symmetric bilinear, quadratic, and derived hermitian theories must be distinguished, and their residue and devissage results must be specified separately. No general dyadic theorem or counterexample is asserted here.

2. Witt groups and mod-2 coefficients

For a scheme X, let $\mathbf{W}(X)$ denote the Witt spectrum and

$$W_n(X) := \pi_n(\mathbf{W}(X)).$$

Write

$$\mathbf{W}(X; \mathbb{Z}/2) := \mathbf{W}(X) \wedge \mathbf{S}/2, \qquad W_n(X; \mathbb{Z}/2) := \pi_n(\mathbf{W}(X; \mathbb{Z}/2)).$$

The mod-2 Bockstein sequence is

$$\cdots \longrightarrow W_n(X) \xrightarrow{2} W_n(X) \longrightarrow W_n(X; \mathbb{Z}/2) \xrightarrow{\partial} W_{n-1}(X) \longrightarrow \cdots . \tag{2}$$

The Gersten–Witt complex with mod-2 coefficients is obtained from the same spectrum-level support filtration, followed by the mod-2 cofiber; it is not the derived tensor product of the integral complex of abelian groups with $\mathbb{Z}/2$.

3. The ring and the self-dual complex

Let

$$R := \Big(\mathbb{Z}_{(3)}[x, u, v, t_1, t_2]/(uv - x^2 - 3)\Big)_{\mathfrak{n}}, \qquad \mathfrak{n} = (3, x, u, v, t_1, t_2). \tag{3}$$

Then R is a regular local ring and

$$\dim R = 1 + 5 - 1 = 5.$$

Put

$$\mathfrak{p} = (3, t_1, t_2) \subset R. \tag{4}$$

The prime $\mathfrak{p}$ has height 3, and

$$R/\mathfrak{p} \cong \mathbb{F}_3[x, u, v]/(uv - x^2) =: S_0. \tag{5}$$

Thus $\kappa(\mathfrak{p}) = \mathrm{Frac}(S_0)$ is the function field of the A_1-quadric cone.

Define

$$A = \begin{pmatrix} x & u \\ v & x \end{pmatrix}, \qquad B = \begin{pmatrix} x & -u \\ -v & x \end{pmatrix}, \qquad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{6}$$

Using $uv - x^2 = 3$, one checks

$$AB = BA = -3I_2, \qquad SA^TS^{-1} = A. \tag{7}$$

Consequently A is self-adjoint for the involution

$$\iota(M) = SM^TS^{-1}$$

on $M_2(R)$.

Consider the two-term complex

$$C_A : \quad 0 \longrightarrow R^2 \xrightarrow{A} R^2 \longrightarrow 0 \tag{8}$$

with the self-duality induced by S.

Lemma 3.1. *The complex C_A is a self-dual perfect complex supported on $V(3)$. After inverting 3, it is contractible. In particular, its generic restriction to $\mathrm{Frac}(R)$ vanishes.*

Proof. The support is the vanishing locus of $\det(A) = x^2 - uv = -3$, hence $V(3)$. Over $R[1/3]$, equation (7) gives

$$A^{-1} = -\frac{1}{3}B,$$

so the two-term complex is contractible. The self-duality is exactly the one induced by the self-adjointness of A in (7). □

4. The residue on the cone

Reducing (7) modulo 3, we obtain

$$\bar{A}\,\bar{B} = \bar{B}\,\bar{A} = 0 \tag{9}$$

over

$$S_0 = \mathbb{F}_3[x, u, v]/(uv - x^2).$$

Thus $(\bar{A}, \bar{B})$ is a matrix factorization of zero over the A_1-quadric cone $uv = x^2$. Let

$$M := \mathrm{coker}(\bar{A}). \tag{10}$$

The module M is the nontrivial rank-one maximal Cohen–Macaulay module over S_0. Its class is the nontrivial element of

$$\mathrm{Cl}(S_0) \cong \mathbb{Z}/2. \tag{11}$$

The self-duality S induces a nondegenerate symmetric form on M. Its determinant is the nontrivial class (11).

Let

$$K := \mathrm{Frac}(S_0) = \kappa(\mathfrak{p}).$$

The residue of C_A at $\mathfrak{p}$ is a class

$$q_\eta \in W_0(K). \tag{12}$$

Lemma 4.1. *The class q_η is a rank-one symmetric form over K. Hence*

$$0 \neq q_\eta \in W_0(K).$$

It is not 2-torsion: its image under the rank map

$$W_0(K) \longrightarrow \mathbb{Z}$$

is 1, so it has infinite order and remains nonzero after inverting 2.

Proof. Over K, the relation $uv = x^2$ holds and A is a rank-one endomorphism of a two-dimensional vector space. The two-term complex $C_A \otimes K$ therefore has one-dimensional cohomology in a single degree. The self-duality S induces a rank-one symmetric form on this cohomology. Its rank is 1. The rank map is nonzero on a rank-one form, and $\mathbb{Z}$ is torsion-free, so the class has infinite order. □

The class q_η extends to the perfect self-dual complex $C_A|_{S_0}$. Therefore its residue at every height-one point of S_0 vanishes. Equivalently, q_η is unramified along the cone S_0.

5. Non-exactness of the Gersten–Witt complex

For a regular local ring R with fraction field K_R, the Gersten–Witt complex begins

$$0 \longrightarrow W_0(R) \longrightarrow W_0(K_R) \longrightarrow \bigoplus_{\operatorname{ht}\mathfrak{q}=1} W_0(\kappa(\mathfrak{q})) \longrightarrow \cdots . \tag{13}$$

The term corresponding to a prime $\mathfrak{q}$ occurs in degree equal to its codimension. Applying the mod-2 cofiber gives the corresponding complex with coefficients $W_0(-;\mathbb{Z}/2)$.

Write $C^\bullet_{\mathrm{GW}}(R;A)$ for this cochain complex, so that

$$C^p_{\mathrm{GW}}(R;A) := \bigoplus_{\operatorname{ht}\mathfrak{q}=p} W_0(\kappa(\mathfrak{q});A), \qquad A \in \{\mathbb{Z}, \mathbb{Z}/2\}.$$

Its cohomology is denoted

$$H^p_{\mathrm{GW}}(R;A) := H^p(C^\bullet_{\mathrm{GW}}(R;A)).$$

The superscript p is the *codimension* or support degree in the Gersten–Witt complex. It is not the degree of a higher Witt group. Thus $W_p(R)$ and $H^p_{\mathrm{GW}}(R)$ are different objects: $W_p(R)$ is a higher Witt group, while $H^p_{\mathrm{GW}}(R)$ is the p-th cohomology group of the Gersten–Witt complex. In particular,

$$H^3_{\mathrm{GW}}(R;\mathbb{Z}/2)$$

means the third cohomology of the mod-2 Gersten–Witt complex, not $W_3(R;\mathbb{Z}/2)$.

The point $\mathfrak{p} = (3, t_1, t_2)$ has codimension 3. The incoming differential at this term comes from the codimension-two points of R, equivalently from the height-one points of

$$S_0 = R/\mathfrak{p}.$$

Each such codimension-two contribution is a residue supported on a proper divisor of S_0. The class q_η is unramified, so it has zero residue at every such divisor.

We use the following standard residue-orthogonality fact for the height-one local rings of the normal surface S_0.

Lemma 5.1 (Residue orthogonality)**.** *Let S_0 be a normal surface with function field K. For each height-one point $\mathfrak{q} \subset S_0$, let*

$$\partial_{\mathfrak{q}} : W_0(K) \longrightarrow W_0(\kappa(\mathfrak{q}))$$

be the residue map. Define the unramified subgroup

$$U(S_0) := \bigcap_{\operatorname{ht}\mathfrak{q}=1} \ker(\partial_{\mathfrak{q}}) \subset W_0(K).$$

Let $R(S_0) \subset W_0(K)$ be the subgroup generated by the images of the ramified lifts

$$\tau_{\mathfrak{q}} : W_0(\kappa(\mathfrak{q})) \longrightarrow W_0(K),$$

obtained by lifting a form over $\kappa(\mathfrak{q})$ to the discrete valuation ring $\mathcal{O}_{S_0,\mathfrak{q}}$ and then extending scalars to K. Then

$$U(S_0) \cap R(S_0) = 0.$$

In particular, a nonzero unramified class is not in $R(S_0)$. The same statement holds after taking the mod-2 cofiber.

Proof. For a height-one point $\mathfrak{q}$, put

$$A_{\mathfrak{q}} := \mathcal{O}_{S_0,\mathfrak{q}}.$$

This is a discrete valuation ring with fraction field K and residue field $\kappa(\mathfrak{q})$. The one-dimensional Gersten–Witt sequence for $A_{\mathfrak{q}}$ gives the exact sequence

$$0 \longrightarrow W_0(A_{\mathfrak{q}}) \longrightarrow W_0(K) \xrightarrow{\partial_{\mathfrak{q}}} W_0(\kappa(\mathfrak{q})) \longrightarrow 0. \tag{14}$$

The lift $\tau_{\mathfrak{q}}$ is the composite

$$W_0(\kappa(\mathfrak{q})) \longrightarrow W_0(A_{\mathfrak{q}}) \longrightarrow W_0(K),$$

where the first map is any lift of a form over the residue field to the DVR; the second is scalar extension. The residue formula for a DVR gives

$$\partial_{\mathfrak{q}} \circ \tau_{\mathfrak{q}} = \mathrm{id}_{W_0(\kappa(\mathfrak{q}))}. \tag{15}$$

Indeed, this is the usual splitting of the second-residue sequence for a discrete valuation ring. In particular, if

$$x = \sum_{\mathfrak{q}} \tau_{\mathfrak{q}}(a_{\mathfrak{q}}) \in R(S_0)$$

is unramified, then applying $\partial_{\mathfrak{q}}$ and using the support of the ramified lifts gives $a_{\mathfrak{q}} = 0$ for every $\mathfrak{q}$. Hence $x = 0$. Thus $U(S_0) \cap R(S_0) = 0$.

The mod-2 statement follows by applying the exact functor $- \wedge \mathbf{S}/2$ to the splitting (15); the residue maps and the lifts are compatible with the Bockstein sequence (2). □

Theorem 5.2. *Let R be the regular local ring* (3) *of dimension* 5*, and let $\mathfrak{p} = (3, t_1, t_2)$. Let*

$$q_\eta \in W_0(\kappa(\mathfrak{p}))$$

be the class (12) *attached to the self-dual matrix factorization* (6)*. Then:*

(i) *$q_\eta \neq 0$ in the integral Witt group $W_0(\kappa(\mathfrak{p}))$;*
(ii) *q_η lies in the kernel of the outgoing residue maps from $\kappa(\mathfrak{p})$ to the height-one points of S_0;*
(iii) *q_η is not in the image of the incoming differential from the codimension-two terms;*
(iv) *consequently the integral Gersten–Witt complex* (13) *for R is not exact at the codimension-three term $\mathfrak{p}$:*

$$H^3_{\mathrm{GW}}(R) \neq 0.$$

Moreover, the same class is nonzero after reduction modulo 2*, and the mod-*2 *Gersten–Witt complex satisfies*

$$H^3_{\mathrm{GW}}(R; \mathbb{Z}/2) \neq 0.$$

Proof. Part (i) is Lemma 4.1. Part (ii) follows because q_η extends to the perfect self-dual complex $C_A|_{S_0}$, so its residue at every height-one point of S_0 vanishes. Part (iii) is the residue-orthogonality Lemma 5.1: the incoming image is the ramified subgroup, while q_η is a nonzero unramified class. Part (iv) follows directly from the definition of exactness of the Gersten–Witt complex. The mod-2 statement follows by applying the mod-2 cofiber to the same support-filtration class; the rank-one form has nonzero rank modulo 2. □

6. Integral versus 2-inverted coefficients

The determinant of the MCM module M is the nontrivial class in $\mathrm{Cl}(S_0) \cong \mathbb{Z}/2$. This does not imply that the Witt class q_η is 2-torsion. The determinant is a class in the Picard group, while q_η has rank 1; its rank gives an infinite-order component in $W_0(K)$.

Thus the construction is naturally an integral construction. The mod-2 statement is a consequence of the integral one. In the 2-inverted theory, the localized complex may become exact because a preimage of q_η can require denominators divisible by 2; this does not make q_η itself zero. Therefore the 2-inverted Gersten–Witt statement does not remove the integral obstruction.

Remark 6.1. The nonfree MCM module over the A_1-cone is the source of the obstruction. In the K-theory counterexample, the analogous role is played by a mod-3 Bott class and a line bundle of order 3. Here the determinant class supplies the nonfree MCM module and the rank-one symmetric form supplies a nonzero integral Witt class.

References

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